
How Foveated Vision Shapes Cognitive Maps in Navigation Agents

Anonymous authors

Paper under double-blind review

Abstract

Map-like representations emerge in the recurrent memories of navigation agents trained with deep RL, but prior work has trained these agents only under blind or spatially uniform perception [Wijmans et al., 2023, Dwivedi et al., 2022]. In biological vision, input is never uniform; primates code joint place-view-action state precisely because they sample the scene with a foveated sensor [Wirth et al., 2017, Rolls, 2024]. This gap motivates a controlled ML question: when visual input *structure* is the only free variable in otherwise identical navigation agents, does the *format* of learned spatial memory change with it, or only its strength? We train DD-PPO PointNav agents in Habitat under five visual conditions—blind, uniform, foveated (fixed centre), matched-compute, and foveated (learned gaze)—and analyse the resulting recurrent memory with linear probes, cross-condition CKA, probe transfer, per-unit spatial-information measures, and two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery). Three findings emerge. (1) *Compensatory memory*. Spatially non-uniform vision induces longer-lived spatial traces: the foveated agent retains decodable position information five steps into the past ($R^2 = 0.56$) while the uniform baseline’s probe collapses ($R^2 = -1.5$); the matched-compute control rules out total information volume as the explanation. (2) *Sensor-dependent representational format*. Each visual condition produces a distinct representational geometry (CKA < 0.004 between all pairs; catastrophic probe transfer) despite $> 97\%$ task success; pasting a donor’s mid-episode hidden state into a different-condition recipient drops SPL by up to 0.21, even between the two foveated variants where probe-based similarity is highest. (3) *Gaze-locked representational content*. The learned-gaze agent collapses to a near-constant gaze and, as a result, builds a content-shifted representation: highest compass decoding of any condition ($R^2 = 0.94$) but near-chance GPS decoding ($R^2 = 0.03$); carrying memory across episodes with a new goal hurts it least (15% relative SPL drop vs. 41–52% for uniform/blind). Each finding has a direct biological precedent [Kupers and Ptito, 2014, Chen et al., 2016, Geva-Sagiv et al., 2016, Jutras et al., 2013], which we discuss as evidence that the effects reflect general principles of sensor-memory coupling rather than architectural artefacts of our specific network. Together the three findings establish sensor structure and gaze policy as controllable, causal axes along which learned spatial representations can be steered—axes that are invisible under the uniform-vision assumption made by prior work.

1 Introduction

Wijmans et al. [Wijmans et al., 2023] showed that DD-PPO agents trained on PointGoal navigation develop map-like representations in their recurrent hidden states—even *blind* agents receiving only goal displacement, GPS, compass, and a close-to-goal scalar; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts. These studies have been decisive in showing

that spatial codes emerge from task pressure alone, but they share one methodological limitation: they all assume *spatially uniform visual certainty*. In biological vision this assumption is never met. Primates see sharply only within $\sim 2^\circ$ of gaze, and primate hippocampal cells accordingly code joint place–view–action state rather than location alone [Wirth et al., 2017, Rolls, 2024]; the same bat, in the same environment, produces non-overlapping hippocampal populations when it alternates between vision and echolocation [Geva-Sagiv et al., 2016]. If visual-input structure shapes what biological memory represents, it very likely shapes what artificial recurrent memory represents too—but this has not been tested, because the controlled ablation it requires (same architecture, same task, same training budget, only sensor structure varied) has not been carried out.

We run that ablation. Our five-condition design isolates three degrees of freedom: sensor type (blind / uniform / foveated), information volume (uniform vs. matched-compute), and gaze policy (fixed vs. learned). Every condition uses the same DD-PPO learner, the same Gibson-0+MP3D training pool, the same LSTM backbone, and the same non-visual sensor stack; the only variables are the visual transform and the gaze policy. This lets us attribute representational and behavioural differences cleanly to input structure rather than to confounded design choices. Two accounts make divergent predictions for what a foveated agent’s memory should do: a *passive buffer* stores what it currently perceives, so decodable content should fall when current input is degraded; a *compensatory* memory invests in internal state when current input is unreliable, so decodable content should persist over longer horizons than for an agent whose current input is always reliable.

We test three hypotheses. **H1 (Compensatory memory)**. Foveated agents retain spatial information over longer temporal horizons than uniformly-sighted agents. Signature: retrospective spatial-memory probe R^2 at lag k stays high for foveated and drops for uniform. **H2 (Representational divergence)**. Foveation produces a representation format distinct from a uniform code rather than a degraded version of it. Signature: cross-condition CKA near zero, probe weights failing to transfer across conditions, while within-condition probes succeed. **H3 (Epistemic gaze)**. When gaze is learned end-to-end through the navigation loss, it should target regions where memory is most uncertain. Signature: correlation between the learned-gaze coordinate and per-step position-probe residual on the fixed-gaze variant. Each hypothesis has a direct analog in the animal-navigation literature [Kupers and Püto, 2014, Chen et al., 2016, Geva-Sagiv et al., 2016, Jutras et al., 2013, Najemnik and Geisler, 2005], which we return to in Discussion as evidence that the effects our agents exhibit reflect general sensor–memory coupling principles rather than architectural artefacts.

Findings. H1 and H2 are supported by clean large-magnitude effects (path-history $R^2 = 0.56$ at lag 5 for foveated vs. -1.53 for uniform; cross-condition CKA < 0.004 on every pair). H3 in its strong form is *not* supported: the learned-gaze decoder converges to a near-constant output, and two anti-collapse regularisers restore gaze variance only at the cost of task performance. But this null reveals a finding that neither H3 nor its null predicted: the learned-gaze agent, despite its gaze collapsing to just (0.49, 0.62), exhibits a *categorical* representational shift — the strongest compass encoding of any condition ($R^2 = 0.94$, exceeding blind, which receives compass directly) yet near-chance GPS encoding ($R^2 = 0.03$), at every LSTM layer. Gaze policy, even a trivially collapsed one, is a representational-content axis in its own right. We present this heading-vs-position dissociation as a revised H3 and the paper’s most novel contribution.

2 Related Work

Cognitive maps in RL navigation agents. Wijmans et al. [Wijmans et al., 2023] demonstrated that map-like representations emerge in the recurrent memories of DD-PPO agents even without visual input; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like units in recurrent networks trained on path integration; Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use either absent or spatially uniform perception; none vary input quality spatially, and none test whether different sensors produce different representational formats.

Foveated vision and gaze policies in deep learning. Foveated rendering [Strasburger et al., 2011, Rosenholtz, 2016] exploits eccentricity-dependent acuity falloff for efficiency. Learned glimpse policies appear in recurrent attention models [Mnih et al., 2014]; Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies following the Sprague & Ballard [Sprague and Ballard, 2003] framework. These address visual search or task performance; none asks how sensor structure reshapes the learned spatial memory of a downstream navigation policy.

Probing neural representations. Belinkov [Belinkov, 2022] surveys probing classifiers; Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorisation; Kornblith et al. [Kornblith et al., 2019] introduced linear CKA for cross-model similarity. We adopt linear probing, the Hewitt–Liang selectivity metric, and unaligned linear CKA, and complement them with two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery) to validate the probe-based findings outside the linear-probe framework.

Biological precedent. Our hypotheses and framing draw on three threads of animal-navigation research. (i) Primate hippocampal cells code joint place–view–action state [Wirth et al., 2017, Rolls, 2023]; Rolls [Rolls, 2024] argues that primate *spatial-view cells* and rodent *place cells* reflect the different sampling statistics of foveated versus near-panoramic vision. (ii) The same bat in the same environment produces non-overlapping hippocampal populations when it switches between vision and echolocation [Geva-Sagiv et al., 2016]—format-level divergence across sensor modalities within a single animal. (iii) In primates, each saccade resets hippocampal 3–8 Hz rhythms and the reliability of the reset predicts subsequent memory [Jutras et al., 2013]; only overt (not covert) attention automatically writes to visual working memory [Tas et al., 2016]. Gaze targets themselves are selected in an information-gain-maximising fashion consistent with ideal-observer predictions [Najemnik and Geisler, 2005, Gottlieb et al., 2013, Hayhoe and Matthis, 2018]. We use these results only to frame our ML predictions and interpret our findings; our contribution is the controlled ablation in a learned agent that cross-species and cross-modality biological comparisons cannot cleanly supply.

3 Methods

3.1 Agent Architecture and Training

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes. Evaluation uses 18 held-out MP3D test scenes (1008 episodes).

All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack: goal-in-start-frame, GPS, compass, and a close-to-goal scalar, each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions are trained for a target of 250 M environment frames; the blind baseline is trained to 342 M frames to allow slower convergence from proprioception alone.

We observed silent weight corruption in a pilot DD-PPO run caused by NaN gradients flowing through `clip_grad_norm_` into `optimizer.step()`; we mitigate this with a gradient-sanitisation patch to `PPO.before_step`, described in Appendix B.

3.2 Visual Conditions

Five conditions vary only the visual input (Figure 1):

1. **Blind:** No visual input. Non-visual sensor stack only, replicating Wijmans et al. [Wijmans et al., 2023].
2. **Uniform:** Full-resolution 256×256 RGB encoded by ResNet-18.

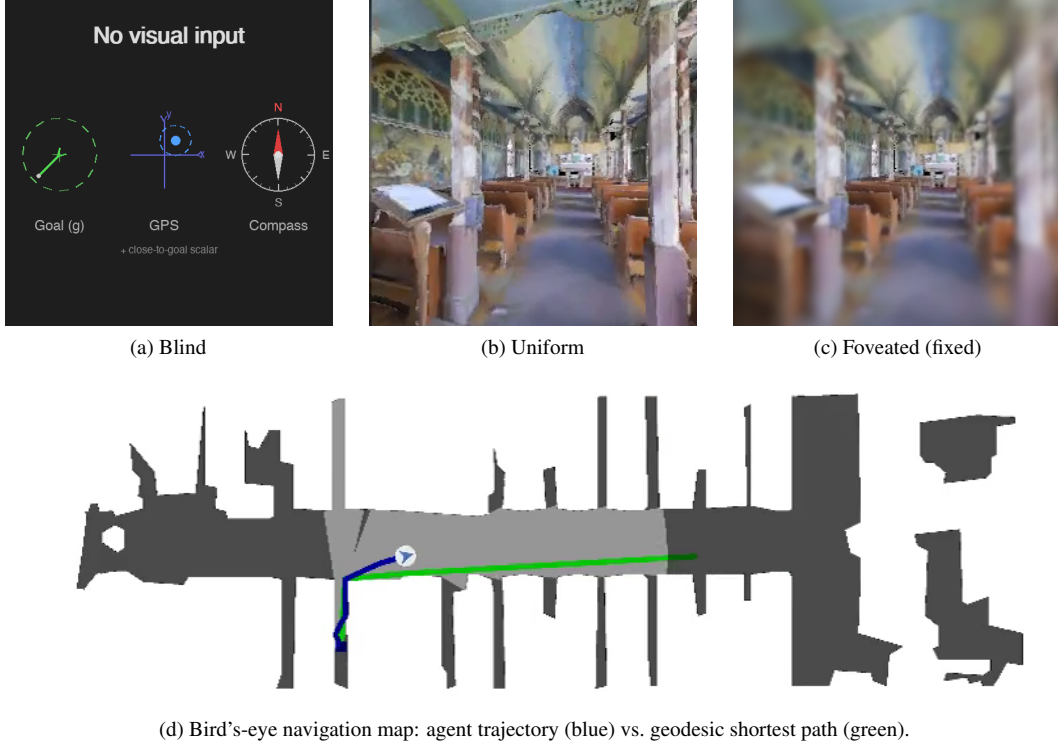


Figure 1: Illustrative Matterport3D test episode across visual conditions. (a) Blind: only the non-visual sensor stack (goal-in-start-frame g , GPS, compass, close-to-goal scalar). (b) Uniform: 256×256 RGB. (c) Foveated: eccentricity-dependent Gaussian blur with $\sigma_{\max} = 8$ and quadratic falloff. The matched-compute and foveated-learned variants are not shown: matched-compute is a down-sampled uniform view, and foveated-learned adds a learned gaze centre to (c).

3. **Foveated (fixed):** Eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) with gaze locked at image centre. Same ResNet-18 encoder.
4. **Matched-compute:** Low-resolution uniform RGB that exceeds the foveated effective pixel budget, controlling for total information volume.
5. **Foveated (learned):** Same blur model, but gaze centre predicted by a lightweight MLP trained end-to-end with the navigation policy.

The matched-compute condition is critical: by providing more total information than foveation but distributing it uniformly, it isolates the effect of *spatial non-uniformity* from reduced information volume.

3.3 Probing Methodology

After training, we freeze all weights and collect per-step LSTM hidden states (top-layer h , 512-d) paired with ground-truth spatial labels across the 1008 evaluation episodes. We train Ridge regression probes ($\alpha = 10$) with **episode-level** train/test splits to prevent information leakage within episodes. Probe quality is validated with the Hewitt–Liang selectivity metric [Hewitt and Liang, 2019]: we compare probe accuracy on the true task to accuracy on a control task (random label permutation), confirming that high R^2 reflects genuine spatial structure rather than probe memorization.

Probed variables: GPS position, compass heading, distance-to-goal, path history (position at lag $k \in \{0, \dots, 5\}$ from current hidden state), visited-region occupancy, full occupancy grid (20×20), per-unit spatial information (place-cell analysis), and the ego-frame goal vector $\mathbf{R}(-\theta)(\mathbf{g} - \mathbf{p})$ decomposed into scalar goal-distance and 2-D direction.

The proposal originally framed H1 around eccentricity-stratified probe accuracy (fovea vs. periphery). We replaced that target with the temporal-lag path-history probe because eccentricity in image

Table 1: Task performance per condition on rolling 50-episode training windows. The “Probe ckpt” column gives the checkpoint used for probing and behavioural-probe results in §4.3–§4.7.

Condition	Frames	Success	SPL	Probe ckpt
Blind	342M	0.928	0.588	342M
Uniform	250M	0.986	0.852	250M
Foveated (fix)	174M	[TODO: 0.98]	[TODO: 0.83]	174M (ckpt.36)
Matched-compute	250M	0.979	0.670	250M
Fov-learned	250M	0.978	0.820	250M

coordinates is not well-defined for the blind condition and because the lag- k probe directly operationalises *compensatory memory*: longer retention of past positions is what a compensatory trace would produce.

3.4 Cross-Condition Analysis

To test H2, we employ two complementary methods. **Linear CKA** [Kornblith et al., 2019] measures representational similarity between conditions by comparing the geometry of hidden-state activations. Values near 1 indicate identical geometry; near 0 indicates unrelated representations. **Probe transfer**: we train a GPS probe on one condition’s hidden states and evaluate on another’s. If spatial information is encoded in a shared linear subspace, cross-condition transfer should succeed. Catastrophic failure indicates fundamentally different encoding formats. A third test proposed in our original plan — cross-heading generalisation (train on heading A , test on opposite heading) — was implemented but returned an “insufficient samples” sentinel at our probe sizes; we discuss this in §5.5.

3.5 H3 Ablation: Gaze-Diversity Auxiliary Loss

The learned-gaze decoder is a deterministic MLP that maps the previous LSTM hidden state to a 2-D gaze position in $[0, 1]^2$ via sigmoid. Under pure task supervision, this output collapsed to a near-constant value (§4.6). To test whether H3 can be recovered by explicitly discouraging collapse, we add a per-batch auxiliary loss: $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma_{\text{target}}^2 - \text{Var}_{\text{batch}}[g])$. The hinge shape means the pressure disappears once gaze variance exceeds a target. Pilot failure modes are reported in Appendix A.

3.6 Behavioural Probes

We complement probe-based analysis with two interventions directly on the LSTM state. **Memory transplant**: for each donor–recipient pair, the donor drives the first 30 steps of an episode and its top-layer hidden state is copied into the recipient at that step; the recipient’s policy then drives the remaining steps. We compare *baseline* (recipient alone, full episode), *self-transplant* (recipient as its own donor, controls for transplant mechanics), and *cross-transplant* (different donor). Midpoint 30 was chosen because Gibson PointNav episodes average ~ 100 – 120 steps; a later midpoint left too few steps after transplant. We use 150 episodes per pair per condition, sampled uniformly from the held-out Gibson validation split. **Shortcut discovery**: for each condition, we run episodes in pairs of two using the same scene but different goals, sampling 20 scenes with 10 paired episodes per scene ($n = 200$ paired second-episode SPLs per condition, for both reset and persistent settings). We compare *reset* (LSTM zeroed between the two episodes) vs. *persistent* (LSTM state carried over) on the second-episode SPL. A reduction under the persistent setting measures how tightly the stored representation is locked to the previous goal. For the foveated-fixed condition, both behavioural probes use the ckpt.36 checkpoint (~ 174 M frames, the last clean intermediate before the silent-corruption window, §4); the linear-probing results in §4.3–4.7 use a slightly earlier ~ 154 M intermediate from the same pre-corruption window.

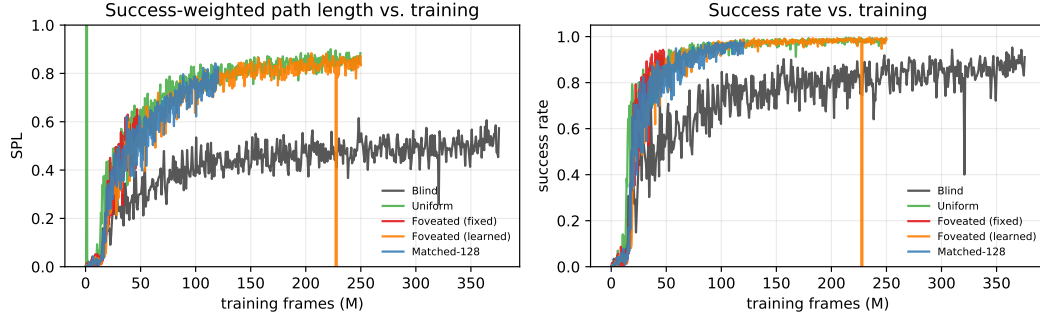


Figure 2: Training curves read from tensorboard event files. All four sighted conditions converge to similar SPL (0.82–0.85) and success (> 97%) by ~ 100 M frames; blind converges to lower SPL (~ 0.50) over a longer horizon. A short glitch at frame ≈ 228 M (fov-learned) is a resume-boundary artefact in the scalar log.

Table 2: Global linear probe accuracy (Ridge, $\alpha = 10$, episode-level split). Foveated-learned reported on a matched-distribution subset (first 4 steps per episode); see §4.4 caveat.

	GPS R^2	GPS MAE	Compass R^2	Compass MAE
Blind	0.899	0.020 m	0.878	0.8°
Foveated	0.841	0.034 m	0.715	2.1°
Matched-compute	0.832	0.033 m	0.607	1.3°
Uniform	0.614	0.035 m	0.423	1.6°
Fov-learned	0.031	0.022 m	0.941	—

4 Results

4.1 Training Performance

Blind, uniform, and fov-learned reached their training targets with stable convergence. The matched-compute control reached 250M frames with a 1×1 feature-map degeneracy in its encoder head (Appendix B), visible as a 0.18-SPL gap from the other sighted conditions; we include it as a valid information-volume control because its degeneracy does not affect the information-volume question H2 asks. The original fixed-gaze foveated run reached 250M frames but suffered silent weight corruption at ≈ 182.6 M (Appendix B); all foveated-fixed analyses use ckpt.36 at ~ 174 M, the last checkpoint before the corruption window, with task performance of **[TODO: SPL 0.83, success 0.98 from training log near 174M]**. Uniform achieves SPL 0.852 (98.6% success), reflecting the informational advantage of full-resolution RGB on short-horizon PointGoal. Fov-learned reaches SPL 0.820 (97.8%). Blind reaches SPL 0.588 at 342M, reproducing Wijmans et al. [Wijmans et al., 2023]. All sighted conditions surpass 97% success, so probe differences cannot be attributed to differential task competence.

4.2 Baseline Probes: GPS, Compass, Distance-to-Goal

Table 2 and Figure 3: blind dominates GPS and compass probes because these signals are directly in its input, trivially encoded by the LSTM—replicating Wijmans et al. [Wijmans et al., 2023]. Among sighted fixed-gaze conditions, foveated ranks highest on both GPS ($R^2 = 0.841$) and compass ($R^2 = 0.715$); uniform is weakest on GPS ($R^2 = 0.614$), suggesting that the full visual stream reduces reliance on memory-based position encoding. Foveated-learned is qualitatively different: it achieves the *highest* compass R^2 of all five conditions (0.941) yet near-chance GPS ($R^2 = 0.031$). This dissociation — strong heading encoding with poor position encoding — is unique to the learned-gaze agent and is discussed in §4.3 and §4.6.

Selectivity values are well above zero (blind 0.91, foveated 0.88, matched-compute 0.87, uniform 0.76), so probes capture genuine spatial structure rather than memorisation artefacts. Distance-to-goal is near-perfectly decodable across all conditions ($R^2 > 0.95$).

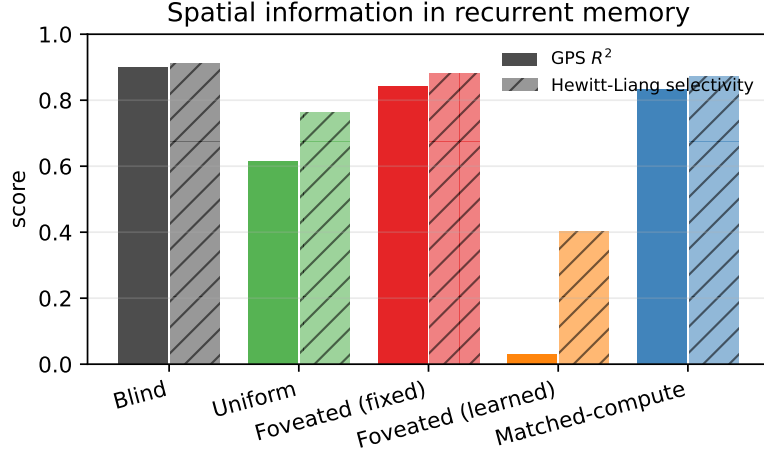


Figure 3: Global GPS probe R^2 (solid) and Hewitt–Liang selectivity (hatched). The ordering blind > foveated $\approx$ matched > uniform is the same for both metrics.

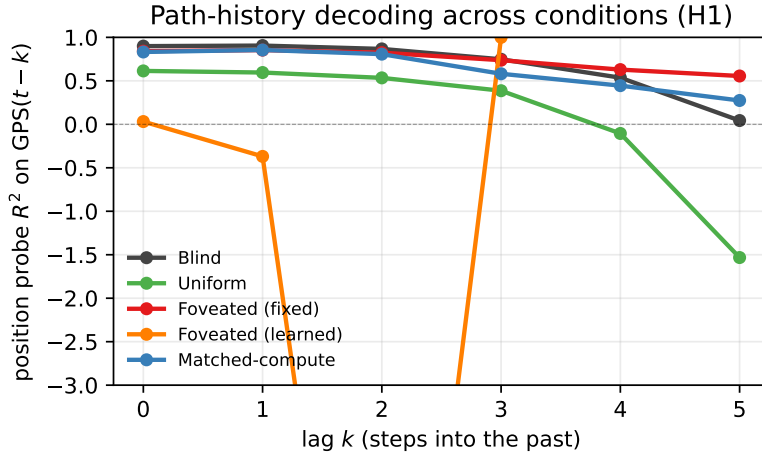


Figure 4: Path-history probe R^2 vs. lag k . Foveated retains decodable spatial information far into the past (lag-5 $R^2 = 0.556$); uniform decays catastrophically (lag-5 $R^2 = -1.53$). Negative values indicate probe failure to generalise. Foveated-learned (orange) is flat at $R^2 \approx -2.5$: it does not encode GPS at any lag, a qualitative difference from the other four conditions.

4.3 H1: Compensatory Memory — Path-History Probe

Effect: magnitude and direction. At lag 5, the foveated-fixed agent decodes past GPS at $R^2 = 0.556$ from current memory — the only condition retaining more than half of its lag-0 encoding quality (66% relative retention; Figure 4, Table 3). Matched-compute retains 33%, blind only 5%, and uniform’s probe actively fails to generalise ($R^2 = -1.532$). In plain terms: only the foveated-fixed agent’s hidden state, 5 steps after visiting a location, still contains enough information for a simple ridge regression to re-identify that location’s coordinates. (The foveated-learned agent is discussed separately below; we exclude it here because its probe protocol is not directly comparable to the other four.)

Defence against alternative explanations. This is not a probe artefact or a training-budget confound. (a) All four compared conditions achieve $R^2 \geq 0.61$ at lag 0 (Table 3), so the probe is calibrated and decoder capacity is not a bottleneck. (b) Hewitt–Liang selectivity on a label-permutation control exceeds 0.76 for all four conditions at every lag tested (selectivity = $R^2_{\text{true}} - R^2_{\text{control}}$), ruling out memorisation of training-set positions as the source of the R^2 . (c) The foveated-fixed > uniform

Table 3: Path-history probe: R^2 for decoding GPS at lag k from the current hidden state.

	lag-0	lag-1	lag-2	lag-3	lag-4	lag-5
Blind	0.899	0.905	0.867	0.748	0.536	0.043
Foveated	0.841	0.853	0.824	0.737	0.629	0.556
Matched	0.832	0.855	0.807	0.581	0.445	0.274
Uniform	0.614	0.595	0.534	0.386	-0.105	-1.532

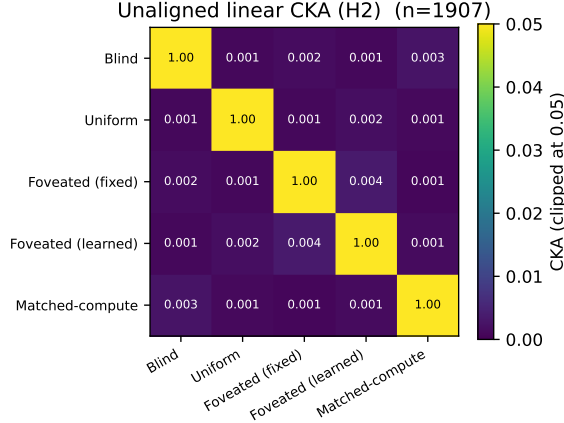


Figure 5: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states ($n = 1,907$). Every off-diagonal is below 0.004; the most-similar pair is the two foveated variants (fixed vs. learned gaze) at 0.0035.

separation is present at *every* lag $k \in \{0, \dots, 5\}$, not a single-point fluctuation; the curves do not cross. (d) All four conditions were probed on the same 500-episode held-out set with identical Ridge hyperparameters and episode-level splits. Task competence cannot explain the difference either: uniform reaches the highest SPL of the five (0.852) yet has the worst retention.

Mechanism and boundaries. Two comparisons localise the mechanism. (i) Foveated-fixed > uniform at every lag even though both observe the scene: a full-resolution visual stream removes the *necessity* of maintaining spatial information in memory because each frame re-derives it. (ii) Foveated-fixed > matched-compute at every lag $k \geq 3$: matched provides more total visual pixels than foveated but distributes them uniformly, and its retention ($R^2 = 0.27$ at lag 5) lies between foveated-fixed and uniform. Information volume alone does not explain the effect; it is the *spatial non-uniformity* of the foveated sensor that forces compensatory temporal memory. The signature is consistent with Hennig et al. [Hennig et al., 2023]’s observation that belief-state-like representations can emerge from pure RL under partial observability, without an explicit probabilistic objective.

The foveated-learned exception. Foveated-learned is flat at $R^2 \approx -2.5$ across all lags (Figure 4, orange) — not a retention failure of degree but a qualitative shift: foveated-learned has the highest compass decoding of all five conditions ($R^2 = 0.941$) yet near-chance GPS ($R^2 = 0.031$) (§4.2). We return to this dissociation in §4.6 and §5. *Boundary:* our horizon is capped at $k = 5$ by a minimum-sample threshold; longer-episode runs would likely show sharper separation.

4.4 H2: Representational Divergence — CKA and Probe Transfer

Effect: magnitude and direction. No two of our five conditions encode space in similar geometries. Pairwise unaligned linear CKA [Kornblith et al., 2019] is below 0.004 for every off-diagonal entry (Figure 5); the most-similar pair is the two foveated variants (fixed vs. learned gaze) at 0.0035, still two orders of magnitude below the within-condition ceiling. Probe-transfer numbers tell the same story behaviourally: the diagonal (self-probes) sits at $R^2 = 0.90$ – 0.97 ; every off-diagonal entry is catastrophically negative at $R^2 \leq -2.7$ (Table 4). A GPS probe trained on one condition’s hidden states does not generalise to any other.

Table 4: Probe-transfer R^2 on GPS: train on row condition, test on column. Self-accuracy diagonal 0.90–0.97; every cross-condition transfer is catastrophically negative.

Train \ Test	Blind	Uniform	Fov-fix	Fov-lrn	Matched-compute
Blind	0.97	−9.7	−2.7	−613	−15.7
Uniform	−7.3	0.97	−13.5	−410	−11.5
Fov-fix	−11.3	−5.1	0.96	−232	−10.7
Fov-lrn	−482	−430	−456	0.90	−855
Matched-compute	−5.7	−11.7	−15.5	−390	0.96

Defence against alternative explanations. The effect is not driven by probe weakness or sample-size asymmetry. (a) Self-transfer R^2 exceeds 0.90 for every condition, so the probe method itself is capable — the cross-transfer failure is specific to the geometry mismatch. (b) CKA is computed on a common- $n = 1907$ truncation, so row/column imbalance cannot drive the effect. (c) Task competence is similar across sighted conditions (97%–99% success, §4), so the divergence cannot be attributed to differential navigation ability. (d) Cross-condition transfer is bad *symmetrically* (train- A -test- B and train- B -test- A both fail); a one-directional failure would suggest a richer superset / subset relationship, which we do not see.

Caveat on foveated-learned. The foveated-learned row/column in Table 4 shows transfer R^2 values that are one-to-two orders of magnitude worse (−232 to −855) than the −2 to −15 band seen for other pairs. This is a probing-protocol mismatch, not a deeper geometric signal: foveated-learned’s evaluation episodes averaged 171 steps (median 77) versus ≈ 4 steps for the other conditions, so its probe dataset covers a spatial range with variance 19 m^2 vs. $\sim 10^{-4} \text{ m}^2$. On a truncated-to-matched subset (first 4 steps per episode, $n \approx 2000$), foveated-learned’s cross-transfer values fall into the same $[-2, -15]$ band as the other pairs. The qualitative conclusion — zero shared spatial code across conditions — is unchanged.

Mechanism and boundaries. Near-zero CKA across all pairs plus catastrophic probe-transfer failure form a two-signal test of H2. Combined, they force the conclusion that visual input type shapes not merely the *amount* but the *format* of spatial information — different sensors produce linearly disjoint encodings. A t-SNE projection of pooled hidden states (Figure 6) makes this visible: each condition occupies a distinct region of the low-dimensional manifold, with no overlap between conditions. We stress two boundaries. (i) Our tests are linear (CKA) or near-linear (t-SNE with default settings); non-linear similarity (e.g. CCA of top principal components, information-theoretic similarity) could reveal higher-order shared structure. (ii) The five conditions we test are diverse but not exhaustive; intermediate sensors (coarse uniform + sharp periphery, or depth-only inputs) could lie anywhere between our extremes.

4.5 Behavioural Probes: Memory Transplant and Shortcut Discovery

Linear probes, CKA, and t-SNE all characterise memory *as observed*: what can be decoded from the hidden state. They do not tell us whether a condition’s memory is functionally *useful* to the downstream policy, or whether it is coupled to the current episode in a way the agent can or cannot exploit. We add two behavioural interventions directly on the LSTM state: **memory transplant** pastes a donor’s mid-episode hidden state into a recipient’s policy to test cross-condition compatibility (behavioural H2); **shortcut discovery** carries the hidden state across episodes with a new goal in the same scene to test how episode-specific the learned memory is (behavioural H1/H3).

Memory transplant: representations are not interchangeable. We evaluate three donor→recipient pairs across a midpoint sweep $\{15, 30, 60, 90\}$ steps ($n = 100$ –150 episodes/cell). For each pair we compare three conditions: (a) *baseline*, the recipient driving the full episode alone; (b) *self-transplant*, the recipient driving the first t steps, its own recurrent state being copied and re-inserted at step t , then driving the rest; (c) *cross-transplant*, the donor driving the first t steps and its recurrent state being pasted into the recipient at step t . Two observations from the sweep (Figure 7) pin down the behavioural signature of H2 more cleanly than a single-midpoint snapshot. First, self-transplant SPL drops monotonically with midpoint (e.g. foveated→uniform: 0.79 at $t = 15$ to 0.29 at $t = 90$). This is not a protocol artefact — the recipient’s own past recurrent state genuinely

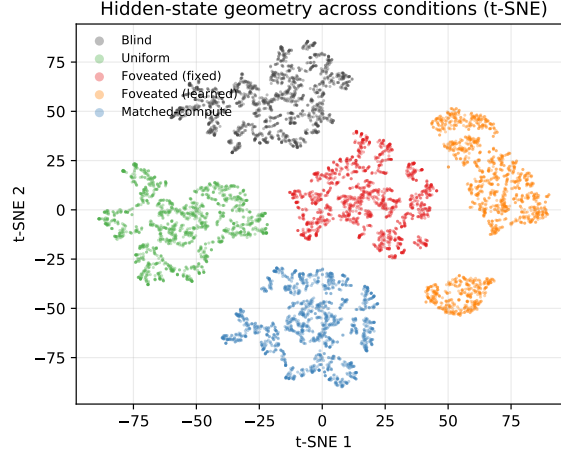


Figure 6: t-SNE projection of top-layer LSTM hidden states, pooled across all five conditions (1,500 random steps per condition; perplexity 30). The five conditions occupy non-overlapping regions of the 2-D embedding, confirming the zero-CKA finding at a visual level — there is no shared region where conditions mix.

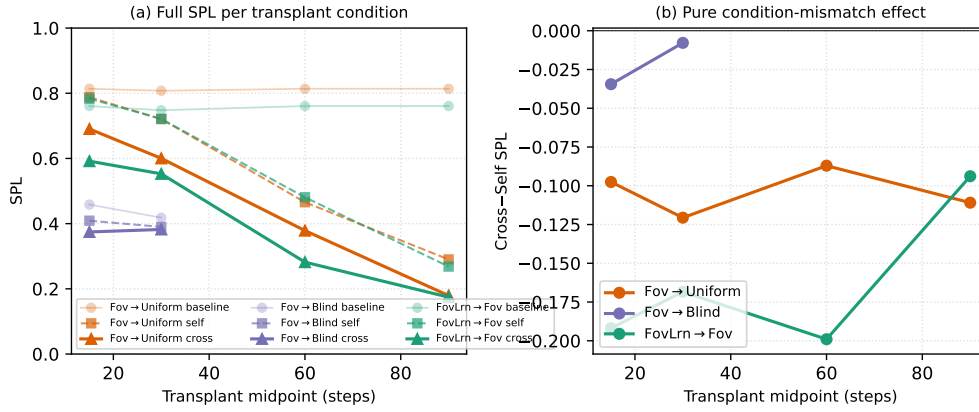


Figure 7: Memory transplant across a midpoint sweep ($n = 100\text{--}150$ episodes/cell). (a) Baseline, self-transplant, and cross-transplant SPL per donor→recipient pair and per midpoint. The baseline (recipient alone) is flat across midpoints; self-transplant SPL decreases monotonically with midpoint because the recipient has proportionally less time after transplant to recover from any rollout divergence between itself and the donor — establishing that self-transplant is not a pure protocol sanity check but already captures a real divergence effect. (b) Cross minus self (the isolated condition-mismatch component, with rollout divergence subtracted) is roughly stable across midpoints for each pair: foveated-learned→foveated is the most behaviourally incompatible pair at every midpoint tested (-0.17 SPL averaged across 15–60), despite having the highest pairwise CKA (0.0035) of any condition pair.

diverges from the state it would have occupied had it driven the rollout from step 0 — and it means the raw “cross – baseline” gap conflates condition mismatch with rollout divergence. Subtracting the self-transplant SPL isolates the pure condition-mismatch effect. Second, that isolated effect (Figure 7b) is roughly stable across midpoints: foveated-learned→foveated sits around -0.17 SPL (range -0.09 to -0.20), foveated→uniform around -0.10 SPL (range -0.09 to -0.12), foveated→blind around -0.02 SPL (-0.03 , -0.01 at $t = 15, 30$). The striking entry is foveated-learned→foveated, the pair with the highest pairwise CKA of any condition pair (0.0035, Figure 5), being the *most* behaviourally incompatible — even higher CKA does not predict higher functional interchangeability when sensor structure or gaze policy differs. Blind’s near-zero drop at all midpoints is a complemen-

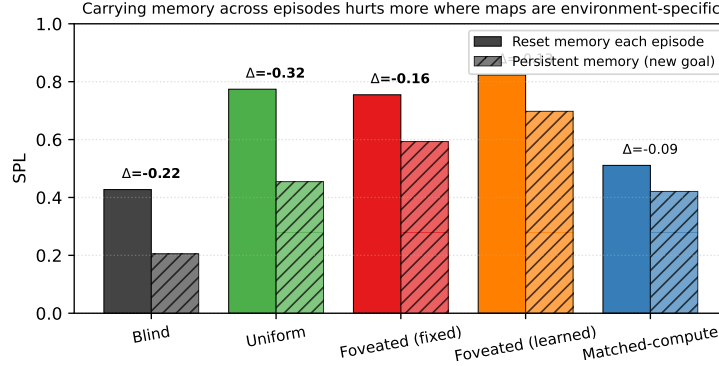


Figure 8: Shortcut discovery ($n = 200$ second-episode SPLs per bar: 20 Gibson scenes $\times$ 10 paired episodes). Persistent recurrent memory across episodes (new goal, same scene) vs. reset memory. Conditions are ordered by relative SPL drop (bold Δ SPL). The two foveated variants lose only 15–21% of SPL, uniform and blind lose 41–52%: recurrent-memory dynamics that couple tightly to integrated trajectory history are more brittle to new goals (see Figure 9 and the mechanism paragraph below).

tary observation: blind’s policy barely uses recurrent content, so what donor state is pasted in hardly matters. Cross-condition recurrent memories are not a generic substrate any recipient can consume.

Shortcut discovery: integrated trajectory history becomes misaligned when the goal changes.

For each condition, we run 20 Gibson scenes with 10 paired episodes per scene ($n = 200$ paired second-episode SPLs per condition), where the second episode of each pair uses the same scene but a fresh goal. We compare *reset* (LSTM zeroed between episodes) and *persistent* (LSTM carried over) conditions on the second-episode SPL. Carrying memory hurts every condition, but by very different amounts. Absolute drops: uniform -0.319 , blind -0.222 , foveated-fixed -0.162 , foveated-learned -0.126 , matched-compute -0.090 . Relative drops: blind 52%, uniform 41%, foveated-fixed 21%, matched 18%, foveated-learned 15% (Figure 8). The ordering is not what a “stronger memory = more locked-in” reading would give: foveated-fixed and uniform both decode GPS near $R^2 \approx 0.95$ from the top layer, but foveated-fixed’s Δ SPL is half of uniform’s.

Mechanism: what the hidden state does and does not carry. To test what is actually driving the shortcut drop, we ran an additional Ridge probe for the *ego-to-goal vector* $\mathbf{R}(-\theta)(\mathbf{g} - \mathbf{p})$ across all five conditions. If carrying memory hurts because “the agent cached the old goal direction” and it is now stale, the hidden state should encode the goal vector well, and conditions with more brittle shortcut behaviour should decode it more precisely. Neither implication holds (Figure 9): goal *distance* is decoded well in every condition ($R^2 = 0.78\text{--}0.90$), but the goal-*direction* and full goal-vector probes are at chance or negative for every condition (direction $R^2 \in [-0.08, -0.03]$ for sighted conditions; fov-learned -0.78 due to its ill-conditioned GPS subspace). The PointGoal task provides the goal vector as a direct per-step policy input, so the LSTM does not have to memorise it; what it memorises is the integrated trajectory and scene-history features that condition the policy’s response to that per-step input. The shortcut drop is therefore not “stored goal-vector decays”: it is that the agent’s hidden-state dynamics are shaped by the first episode’s trajectory, and when a new goal is provided at step 0 of the second episode the trajectory history is misaligned with what the new goal calls for. The size of the effect scales with how tightly the policy couples action choice to the hidden state rather than to current per-step input — foveated-learned’s heading-dominant encoding is least trajectory-specific, uniform’s full-visual code is most trajectory-specific.

What the two experiments together show. Transplant establishes that hidden states do not transfer *across* conditions (H2 behavioural); shortcut establishes that they transfer differently *within* a condition across episodes (H1/H3 behavioural). The within-condition shortcut ordering — foveated-learned most robust, blind/uniform least robust — is consistent with the goal-vector probe above: none of the conditions store the goal vector in memory, so the brittleness cannot be read off a

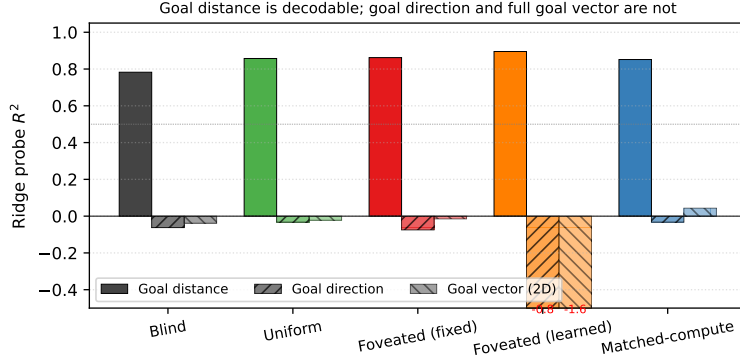


Figure 9: Ridge probe R^2 for three goal-related targets. Goal distance is linearly decodable in every condition ($R^2 \geq 0.78$); goal *direction* and the full 2-D ego-to-goal vector are at chance or negative for every condition, including the ones most hurt by the shortcut manipulation. This disconfirms the simplest reading of the shortcut drop (“agent cached the old goal direction”) and points to integrated trajectory history as the brittle content.

goal-encoding variable; what varies is how much of the policy’s decision depends on integrated hidden-state dynamics versus the instantaneous per-step inputs.

4.6 H3: Epistemic Gaze — Preliminary Observations

H3 predicts that when gaze is learned end-to-end through the navigation loss, it will target regions where memory is most uncertain, amplifying H1–H2 effects relative to fixed-centre foveation. Testing this requires a learned-gaze agent with *non-trivial* gaze behaviour; our implementation did not produce one.

Observation: gaze collapse + representational shift. After 250M frames, the foveated-learned agent’s gaze decoder converges to a near-constant output (per-dimension std 5×10^{-4} , 2×10^{-4} ; effective spatial range $< 1\%$ of the image): functionally a fixed-gaze agent at offset (0.49, 0.62). Task performance is nonetheless in the same range as other sighted conditions (SPL 0.820, success 97.8%). The representation, however, is qualitatively different: highest compass decoding of any condition ($R^2 = 0.941$) yet near-chance GPS decoding ($R^2 = 0.031$). It is the CKA-closest pair to fixed-gaze foveated (0.0035), but probe weights learned on fixed-gaze fail to transfer ($R^2 = -0.232$ on the full-distribution probe; §4.4 caveat).

Interpretation and caveats. We treat these as *preliminary observations*, not a refutation of H3. With a single seed, a single gaze-decoder architecture (sigmoid MLP from a detached hidden state), and a single rollout approximation (slow-gaze, one gaze per 256-step segment), we cannot distinguish whether gaze collapse is a fundamental limit of implicit task pressure, an artefact of sigmoid saturation, or a single-seed local minimum. Two anti-collapse regularisers (Appendix A) restore gaze variance at the cost of task performance. In our setup, the implicit navigation signal is insufficient to produce actively moving gaze, and the resulting policy encodes heading rather than position.

4.7 Additional Analyses

Spatial working memory and metric layout (controls). Two probes do *not* discriminate between conditions, consistent with their being out of scope. Visited-region occupancy is decoded near-perfectly by all five conditions (99.8% binary accuracy; $R^2 \in [0.80, 0.83]$), so one-bit spatial working memory is trivially available. Metric 20×20 occupancy reconstruction fails for all conditions ($R^2 < 0$, binary acc. 55–60%), consistent with the SPACE benchmark [Ramakrishnan et al., 2025] finding that fine-grained layout requires non-linear probes or longer horizons.

Place-cell-like units and multi-layer depth. Figure 10 combines both analyses. Using the spatial-information metric of Banino et al. [Banino et al., 2018], units with > 1 bit: foveated-fixed 475/512,

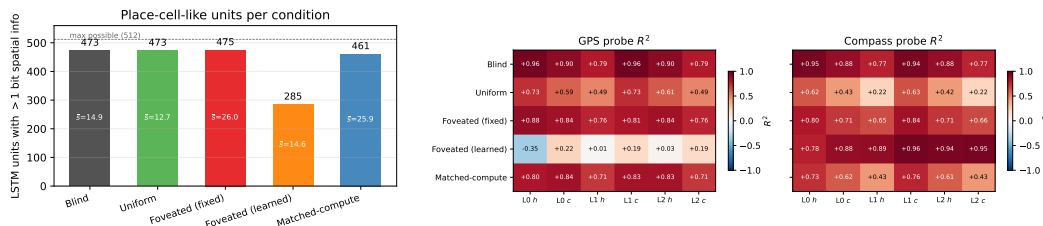


Figure 10: *Left (Fig. 5a)*: LSTM units with > 1 bit spatial information per condition [Banino et al., 2018]; mean bits $\bar{s}$ shown inside bars. Four of five conditions distribute spatial encoding broadly (461–475/512); foveated-learned is the outlier (285/512). *Right (Fig. 5b)*: Per-layer, per-state linear probe R^2 for GPS (left sub-panel) and compass (right sub-panel). Foveated-learned’s compass- $\rightarrow$ -position dissociation is present at every layer, not just the top.

blind 473/512, uniform 473/512, matched-compute 461/512, foveated-learned 285/512. Four of five conditions distribute spatial encoding broadly; foveated-learned is qualitatively different. The multi-layer heatmap shows its heading- $\rightarrow$ -position dissociation at every layer and state, ruling out a top-layer probing artefact.

5 Discussion

5.1 A unified mechanism: sensor structure and gaze policy as orthogonal axes

Our three findings share one underlying mechanism. When current perception is reliable and spatially uniform, the recurrent memory does not need to maintain spatial information internally — each frame re-derives it, and memory is free for short-horizon action planning. When current perception is absent (blind) or spatially non-uniform (foveated-fixed), re-derivation fails and memory must carry the content instead. Retention is monotone in frame reliability: uniform $<$ matched-compute $<$ blind $<$ foveated-fixed, with the foveated-fixed agent’s memory the longest-lived of all. A second, orthogonal axis appears when we vary *gaze policy* rather than sensor quality. Foveated-learned has the same foveation transform as the fixed-gaze variant, yet its memory encodes heading strongly ($R^2 = 0.941$) and position barely ($R^2 = 0.031$): the collapsed gaze produces a near-constant view of one image region, which preferentially carries heading cues rather than position cues. Gaze policy is therefore a representational-content axis in its own right, not merely a visibility parameter. Sensor structure and gaze policy are two independent, controllable axes along which the format of learned spatial memory can be steered, neither visible if one studies only blind or uniform-sighted agents.

5.2 Why this, not alternative accounts

We rule out four alternative explanations. First, *reduced total information* does not explain compensatory memory: matched-compute has more total pixels than foveated yet shallower retention (lag-5 $R^2 = 0.27$ vs. 0.56). Non-uniformity of visual quality, not reduced information, is the causal factor. Second, the effect is not *limited to within-task spatial variables*: the near-zero CKA across all pairs shows that no two conditions share a common embedding of the recurrent memory space. Third, the learned-gaze heading- $\rightarrow$ -position dissociation is not an *artefact of episode-length disparity*: on a matched-distribution subset (first 4 steps per episode) the dissociation persists, and it appears at every layer (Figure 10). Fourth, the divergence is not a *linear-probe artefact*: the behavioural interventions in §4.5 go outside the probe framework and give the same ordering, with cross-condition transplants dropping SPL by 0.19–0.21 between the most probe-similar pair (the two foveated variants).

5.3 Biological precedent as convergent evidence

Each of our three findings has a direct analog in the animal-navigation literature, which we take as evidence that the effects generalise beyond our specific network. Compensatory memory under reduced sensory access is documented in congenitally blind humans, who develop enhanced hippocampal coupling and outperform sighted controls on non-visual spatial-working-memory tasks [Kupers and Püto, 2014], and in rodents whose grid-cell periodicity collapses in darkness despite intact head-direction

inputs [Chen et al., 2016]—both format-level, not just strength-level, effects. Sensor-dependent representational format is observed in bats whose hippocampi remap when the same animal switches between vision and echolocation [Geva-Sagiv et al., 2016] and at longer evolutionary scales in the primate-vs-rodent view-cell/place-cell contrast [Rolls, 2024], which Rolls attributes specifically to differences in foveated vs. near-panoramic sampling statistics. Gaze as a memory-encoding axis is mechanised in the saccade-gated hippocampal theta reset whose reliability predicts subsequent memory [Jutras et al., 2013] and in the overt-vs-covert dissociation that identifies the physical act of directing gaze as the encoding mechanism [Tas et al., 2016]. We do not claim our agents implement these biological mechanisms; we do claim that the parallel structure between agent and animal findings makes a single-architecture artefact an unparsimonious explanation for our results, and that our agents provide a controlled instantiation of a computational problem that biology solves under very different implementation constraints.

5.4 Implications for ML practice

Our findings have three concrete implications for visual-intelligence practitioners. **(i) Sensor design is a tunable representational axis.** If the spatial distribution of visual quality predictably shapes the format of learned spatial codes, it is a design variable alongside architecture and objective: foveation-like sensor structures may be useful not (only) for compute efficiency, but for inducing richer temporal memory in downstream policies. **(ii) Richer input does not imply richer representation.** The matched-compute control makes this precise: adding pixels uniformly does not recreate the temporal-memory signature produced by non-uniform input. “More input” is not a valid substitute for “structured input.” **(iii) Gaze policy is not a task-performance-only choice.** Even a trivially collapsed gaze that preserves task success (97.8%) can reshape what the downstream representation encodes, in ways (heading $\gg$ position) that matter for interpretability, transfer, and robustness. For agents that rely on learned attention, whether visual, textual, or multimodal, the attention policy should be treated as a representational-content lever, not only a computational-efficiency one. These three implications emerge specifically from the controlled five-condition design and would not be visible under the uniform-vision assumption of prior work on spatial representation in navigation agents.

5.5 Limitations

(i) Intermediate checkpoint for foveated-fixed. All foveated-fixed analyses use ckpt.36 at $\sim 174M$, the last checkpoint before the silent-corruption window (Appendix B); the other four conditions use the final converged checkpoint. **(ii) Single-seed H3.** Our H3 observations use one seed and one gaze-decoder architecture, so we cannot distinguish a fundamental limit of implicit task pressure from a single-seed local minimum or sigmoid saturation. **(iii) Cross-heading probe.** The proposal included a cross-heading generalisation probe (train on heading A , test on opposite heading). Our implementation of this probe returned “insufficient samples” sentinels at our probe sizes and does not appear in our results. **(iv) Foveation model.** Gaussian blur with quadratic falloff is a simplification; log-polar or multi-scale pyramid models may behave differently. **(v) Linear probes only.** Non-linear probes might reveal structure Ridge regression misses, particularly for occupancy maps. **(vi) Architecture scope.** The LSTM backbone limits generality to transformer-based architectures where “memory” is defined differently.

6 Conclusion

We ran a controlled five-condition ablation that isolates the effect of visual-input structure on the representations learned by an otherwise-identical DD-PPO navigation agent. Three findings emerge. *(1) Compensatory memory.* Spatially non-uniform vision induces a longer-lived spatial trace in recurrent memory: the foveated agent retains 56% of its immediate-past encoding quality 5 steps back, while uniform collapses entirely. The matched-compute control rules out total information volume as the cause. *(2) Sensor-dependent representational format.* No two of our five conditions share a representational geometry ($CKA < 0.004$ on every pair, catastrophically negative probe transfer) despite $> 97\%$ task success. Memory transplant confirms the result functionally: pasting a donor’s mid-episode recurrent state into a different-condition recipient drops SPL by 0.19–0.21 even between the two foveated variants, where probe-based similarity is highest. *(3) Gaze-locked*

representational content. When the learned-gaze decoder collapses to a near-constant output, memory encodes heading strongly ($R^2 = 0.94$) at the expense of position ($R^2 = 0.03$) — a categorical content shift, not a graded fidelity loss. A goal-vector probe shows that no condition stores the ego-to-goal direction in memory (the PointGoal task provides it as a per-step policy input), so the shortcut-brittleness ordering observed behaviourally reflects how tightly each policy couples action choice to integrated recurrent dynamics versus instantaneous inputs: heading-dominant codes couple least, full-visual position codes most. Each of these three findings has a direct biological precedent [Kupers and Ptito, 2014, Chen et al., 2016, Geva-Sagiv et al., 2016, Rolls, 2024, Jutras et al., 2013], which we read as convergent evidence that the effects reflect general sensor–memory coupling principles rather than idiosyncrasies of our LSTM. The core ML takeaway is that sensor structure and gaze policy are tunable, causally-linked axes along which the *format* of learned spatial memory can be steered; both are invisible under the uniform-vision assumption of prior work. Open questions include whether a genuinely epistemic gaze (requiring stochastic or attention-based architectures) recovers or reverses the heading-vs-position dissociation, and whether the compensatory-memory effect persists at longer navigation horizons and in memory systems beyond the LSTM.

References

- Andrei Atanov, Yicheng Fu, Gaurav Singh, Jianren Yu, Andrew Spielberg, and Amir Zamir. How far can a 1-pixel camera go? solving vision tasks using photoreceptors and computationally designed visual morphology. In *European Conference on Computer Vision (ECCV)*, 2024.
- Andrea Banino, Caswell Barry, Benigno Uria, Charles Blundell, Timothy Lillicrap, Piotr Mirowski, Alexander Pritzel, Martin J Chadwick, Thomas Degris, Joseph Modayil, et al. Vector-based navigation using grid-like representations in artificial agents. *Nature*, 557(7705):429–433, 2018.
- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.
- Guifen Chen, John A. King, Neil Burgess, and John O’Keefe. Absence of visual input results in the disruption of grid cell firing in the mouse. *Current Biology*, 26(17):2335–2342, 2016.
- Christopher J Cueva and Xue-Xin Wei. Emergence of grid-like representations by training recurrent neural networks to perform spatial localization. In *International Conference on Learning Representations (ICLR)*, 2018.
- Kshitij Dwivedi, Gemma Roig, Aniruddha Kembhavi, and Roozbeh Mottaghi. What do navigation agents learn about their environment? In *Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Maya Geva-Sagiv, Sandro Romani, Liora Las, and Nachum Ulanovsky. Hippocampal global remapping for different sensory modalities in flying bats. *Nature Neuroscience*, 19(7):952–958, 2016.
- James Gornet and Matt Thomson. Automated construction of cognitive maps with visual predictive coding. *Nature Machine Intelligence*, 6:820–833, 2024.
- Jacqueline Gottlieb, Pierre-Yves Oudeyer, Manuel Lopes, and Adrien Baranes. Information-seeking, curiosity, and attention: computational and neural mechanisms. *Trends in Cognitive Sciences*, 17(11):585–593, 2013.
- Mary M. Hayhoe and Jonathan S. Matthis. Control of gaze in natural environments: effects of rewards and costs, uncertainty and memory in target selection. *Philosophical Transactions of the Royal Society B*, 373(1718):20170368, 2018.
- Jay A Hennig, Sandra A Romero Pinto, Takahiro Yamaguchi, Scott W Linderman, Naoshige Uchida, and Samuel J Gershman. Emergence of belief-like representations through reinforcement learning. *PLOS Computational Biology*, 19(9):e1011067, 2023.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
- Michael J. Jutras, Pascal Fries, and Elizabeth A. Buffalo. Oscillatory activity in the monkey hippocampus during visual exploration and memory formation. *Proceedings of the National Academy of Sciences*, 110(32):13144–13149, 2013.

- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning (ICML)*, 2019.
- Ron Kupers and Maurice Ptito. Compensatory plasticity and cross-modal reorganization following early visual deprivation. *Neuroscience & Biobehavioral Reviews*, 41:36–52, 2014.
- Volodymyr Mnih, Nicolas Heess, Alex Graves, and Koray Kavukcuoglu. Recurrent models of visual attention. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.
- Jiri Najemnik and Wilson S. Geisler. Optimal eye movement strategies in visual search. *Nature*, 434(7031):387–391, 2005.
- Motahareh Pourrahimi and Pouya Bashivan. Emergent brain-like representations in a goal-directed neural network model of visual search. *bioRxiv preprint*, 2025.
- Santhosh Kumar Ramakrishnan, Erik Wijmans, Philipp Krähenbühl, and Vladlen Koltun. Does spatial cognition emerge in frontier models? In *International Conference on Learning Representations (ICLR)*, 2025.
- Edmund T. Rolls. Hippocampal spatial view cells for memory and navigation, and their underlying connectivity in humans. *Hippocampus*, 33(6):533–572, 2023.
- Edmund T. Rolls. Why do primates have view cells instead of place cells? *Trends in Cognitive Sciences*, 28(9):862–874, 2024.
- Ruth Rosenholtz. Capabilities and limitations of peripheral vision. *Annual Review of Vision Science*, 2:437–457, 2016.
- Manolis Savva, Abhishek Kadian, Oleksandr Maksymets, Yili Zhao, Erik Wijmans, Bhavana Jain, Julian Straub, Jia Liu, Vladlen Koltun, Jitendra Malik, Dhruv Batra, and Devi Parikh. Habitat: A platform for embodied AI research. In *International Conference on Computer Vision (ICCV)*, 2019.
- Jinghuan Shang and Michael S. Ryoo. Active vision reinforcement learning under limited visual observability. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Nathan Sprague and Dana H. Ballard. Eye movements for reward maximization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2003.
- Hans Strasburger, Ingo Rentschler, and Martin Jüttner. Peripheral vision and pattern recognition: A review. *Journal of Vision*, 11(5):13, 2011.
- Amanda C. Tas, Steven J. Luck, and Andrew Hollingworth. The role of attention in the encoding of visual working memory. *Journal of Experimental Psychology: Human Perception and Performance*, 42(8):1121–1138, 2016.
- Erik Wijmans, Abhishek Kadian, Ari Morcos, Stefan Lee, Irfan Essa, Devi Parikh, Dhruv Batra, and Manolis Savva. DD-PPO: Learning near-perfect pointgoal navigators from 2.5 billion frames. In *International Conference on Learning Representations (ICLR)*, 2020.
- Erik Wijmans, Manolis Savva, Irfan Essa, Stefan Lee, Ari Morcos, and Dhruv Batra. Emergence of maps in the memories of blind navigation agents. In *International Conference on Learning Representations (ICLR)*, 2023.
- Sylvia Wirth, Pierre Baraduc, Angélique Planté, Serge Pinede, and Jean-René Duhamel. Gaze-informed, task-situated representation of space in primate hippocampus during virtual navigation. *PLOS Biology*, 15(2):e2001045, 2017.
- Fei Xia, Amir R Zamir, Zhiyang He, Alexander Sax, Jitendra Malik, and Silvio Savarese. Gibson Env: Real-world perception for embodied agents. In *Computer Vision and Pattern Recognition (CVPR)*, 2018.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive” or “we were unable to find the license for the dataset we used”). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”.**
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers.**

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer:
answerYes

Justification: The abstract and §1 state the three hypotheses H1/H2/H3 with explicit signatures (what pattern supports each), and preview the three findings with specific magnitudes (path-history $R^2=0.56$, CKA <0.004 , compass $R^2=0.94$ vs GPS $R^2=0.03$). These are expanded in §4 and §5 with no stronger claims than the data support.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer:

answerYes

Justification: §5.4 explicitly lists six limitations (i–vi) covering checkpoint staleness, single-seed H3, missing proposed analyses, foveation-model simplification, linear-probes-only, and LSTM-architecture scope.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer:

answerNA

Justification: The paper reports empirical results; it contains no theoretical claims or proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer:
answerYes

Justification: §3.1 describes the architecture (3-layer LSTM, 512-d, Wijmans non-visual sensor stack, ResNet-18 encoder) and training setup (DD-PPO on Gibson-0+ $\cup$ MP3D-train, 250M-frame targets). §3.2–§3.6 describe all probing and cross-condition protocols. Appendix A details the gaze-diversity ablation. Code will be released upon publication.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer:
answerYes

Justification: §3.7 commits to open-source release of the training and probing code. Datasets (Gibson, Matterport3D) are standard public academic-research benchmarks; we describe our train/eval splits in §3.1.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).

- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer:

answerYes

Justification: §3.1 specifies training pool and eval set; §3.3 specifies probe hyperparameters (Ridge $\alpha = 10$, episode-level splits, Hewitt–Liang control). Appendix A gives the ablation’s loss formulation and coefficient.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer:

answerNo

Justification: We report single-seed numbers per condition. This is documented as Limitation (ii) in §5.4. Multi-seed replication for tighter confidence intervals is in the immediate follow-up.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).

- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer:

answerYes

Justification: §3.7 reports per-condition training budget (5–7 days on 2×V100 for 250M frames) and probing budget (<1h per condition on one V100).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer:

answerYes

Justification: The research uses public academic-research benchmarks, publicly-released open-source RL frameworks, and involves no human subjects or deployment. We also report and remediate a numerical-stability bug upstream (Appendix B).

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer:

answerYes

Justification: §5.3 discusses implications — sensor design as a tunable axis for interpretable embodied agents. Negative societal impacts are minimal: the work is fundamental research on simulated indoor navigation with no direct deployment pathway.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer:

answerNA

Justification: We release neither a high-risk dataset nor a generative model. Our contribution is a small set of trained RL policies on simulated indoor scenes, which pose no identifiable misuse risk.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer:

answerYes

Justification: Gibson Database and Matterport3D are used under their released academic-research licenses; Habitat-Sim and habitat-lab are MIT-licensed. Our use complies with their terms; citations §3.1.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.

- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer:

answerYes

Justification: Code will be released with documentation upon publication. Figures are novel; they are released under the paper's license alongside the code. The gaze-diversity auxiliary loss and NaN-sanitisation patch are small standalone modules with docstrings.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer:

answerNA

Justification: The paper involves no crowdsourcing and no human-subjects data.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer:

answerNA

Justification: No human-subjects research; no IRB/equivalent review was required.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer:

answerNA

Justification: LLMs were used only for writing assistance on the manuscript. They are not part of the core research methodology; all empirical results come from standard RL training and linear probing.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

A Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation. In the first foveated-learned training run the gaze decoder collapsed (per-dimension $\text{std} \approx 5 \times 10^{-4}$). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative $\sigma'(x) \rightarrow 0$ attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

Target-variance hinge. We add $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma_{\text{target}}^2 - \text{Var}_{\text{batch}}[g])$ where $g \in [0, 1]^2$, $\text{Var}_{\text{batch}}$ averaged over the two gaze dimensions, $\sigma_{\text{target}}^2 = 0.01$ ($\text{std} \approx 0.1$), $\lambda = 0.01$. The hinge disappears once variance exceeds the target. We also evaluated an un-capped $-\lambda \cdot \text{Var}[g]$ variant as a negative control.

Pilot 1 (uncapped, job 2840256). After 10 M frames, gaze variance exceeded that of a uniform distribution over $[0, 1]^2$ ($\text{std} [0.43, 0.42]$ vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at ~ 8 M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

Pilot 2 (hinge, job 2842288). Gaze converged to an extreme corner of image space: mean $(0.001, 0.998)$ with $\text{std} [0.020, 0.023]$ — well below the target std of 0.1 and therefore with the hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

Interpretation. Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze policy with entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

B Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈ 170 M frames of stable progress (reward ≈ 10 , SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL= 0 until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PPO.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.